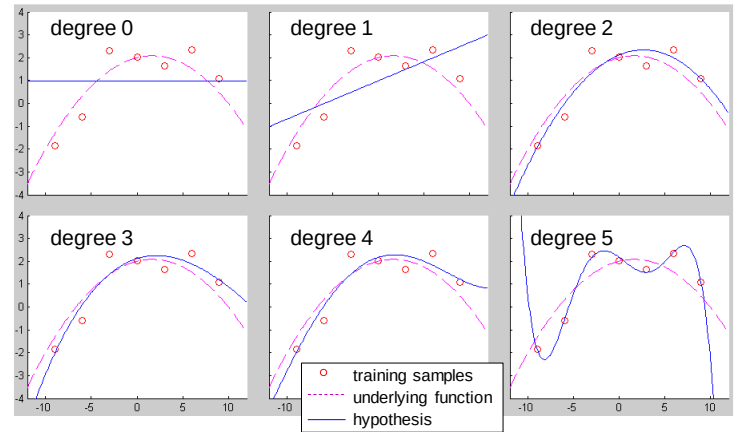


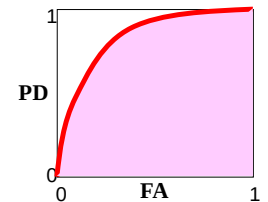
- There are several parts (A, B, ..., G) in this exam. Each part has a separate submission link on E3.
- You need to handwrite your answers on paper, and submit the photographed or scanned image file.
- Be sure to include your name and ID inside each submitted image.
- This is an open-book exam. You can consult the lectures or other books. However, you should do this independently and are not allowed to discuss with each other.

Part A: Supervised Learning [20%]

1. [8%] The figure here is an example of polynomial regression discussed in the class. Use polynomial regression to discuss the bias-variance dilemma:
 - (a) [4%] What are the bias and variance in the problem of polynomial regression?
 - (b) [4%] How do the bias and variance change with the degree of the polynomial?



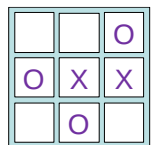
2. [5%] For a standard ROC curve like the one shown here, indicate how the following quantities change as we move along the ROC curve (from bottom-left to top-right): true-positive (TP), false-positive (FP), true-negative (TN), false-negative (FN), recall, sensitivity, precision, specificity, and F1.
3. [3%] In PCA, we project some higher dimensional data to a lower dimensional space. What quantity do we maximize when selecting the basis of projection, assuming that the dimensionality after projection is fixed?
4. [4%] What is the meaning of the diagonal elements of a confusion matrix? How is their sum (the trace) used in classifier evaluation?



Part B: Games [15%]

From this game state of the tic-tac-toe game, and the next player (MAX) is 'X':

- (a) [6%] Draw the 2-ply game tree.
- (b) [5%] Use the following evaluation function to compute values of the leaf nodes in (a):
"the count of the player's unblocked two-in-a-lines, minus the count of the opponent's unblocked two-in-a-lines"



Note: An unblocked two-in-a-line is any line (vertical, horizontal, or diagonal) where two spaces are occupied by the same player and the third space of the line is still empty.

- (c) [4%] Use the minimax rule to determine the action to be taken by X. Note: If there are multiple actions that are equally good, indicate them all.

Part C: Unsupervised Learning [10%]

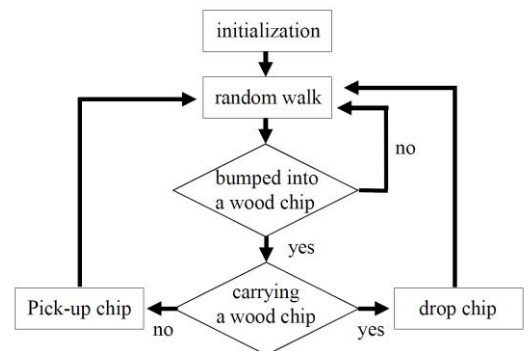
- (a) [5%] Describe the difference of k-medoid clustering from k-means clustering. Also, does k-medoid clustering applicable to relational data? Briefly explain.
- (b) [5%] List the three steps of self-organizing maps, and indicate which of the steps is different from vector quantization. What is the effect of this step?

Part D: Reinforcement Learning [15%]

- (a) [3%] Use your own words to explain what a "policy" is.
- (b) [3%] Why are the state values dependent upon the policy?
- (c) [4%] Policy iteration is an alternating optimization procedure for determining the optimal policy, assuming that the reward functions and transition probabilities are all known. What are the two parts in policy iteration?
- (d) [5%] What is a Q function in reinforcement learning? How do you select the optimal policy given the Q function?

Part E: Evolutionary Computation and Swarm Intelligence [17%]

1. [5%] Describe the two standard genetic operators in GA. Which one leads to new individuals that are similar to the parents?
2. [3%] What is the difference between generational and steady-state evolution?
3. [3%] What is the purpose of a fitness function?
4. [6%] See the flowchart of termite mound building to the right:
 - (a) [3%] According to the flowchart, do the termites use direct or indirect communication?
 - (b) [3%] What part of the flowchart corresponds to "stigmergy" in swarm intelligence? Briefly explain.



Part F: Logical Inference [15%]

1. [7%] An inference rule

given
inferred

 is sound iff: whenever all the conditions in "given" are true, "inferred" is true. We can use truth tables to prove that a given inference rule is sound by enumerating the truth values of all the variables.
 - (a) [4%] Use truth table to prove that Modus Ponens is a sound inference rule.
 - (b) [3%] Use a counter example ("given" is true and "inferred" is false) to show that this rule

$\alpha \Rightarrow \beta$
$\beta \Rightarrow \alpha$

 is not sound.
2. [3%] Translate the following first-order logic sentence into plain English:
$$\forall x \text{ King}(x) \wedge \text{Greedy}(x) \Rightarrow \text{Evil}(x)$$
3. [5%] Explain what a satisfiability problem in propositional logic is; give a solution to satisfy this:
$$(\alpha \wedge \beta) \vee (\neg \alpha \wedge \neg \beta)$$

Part G: Natural Language Processing [8%]

- (a) [4%] When we want to compare documents according to the words in them, why do we need the IDF part (as in TF-IDF)?
- (b) [4%] We mentioned several preprocessing tasks in natural language processing. Name one task that does not consider the context, and one task that needs to consider the context.

Name: _____ ID: _____

Prob#	1-4	5-7	8-12	13-16	Total
Score					

1. [6%] Describe the main difference between generative and discriminative classification models, and name at least one example each.

2. [8%] This is the cost function, with regularization, for estimating the coefficients a_k , such that a function given by the linear combination of a set of kernels, $\phi_k(x)$, approximates a given set of (x_i, y_i) pairs.

$$E = \sum_i \left| y_i - \sum_{k=1}^d a_k \phi_k(x_i) \right|^2 + \lambda \sum_{k=1}^d a_k^2$$

(a) [3%] Explain how the adjustable parameter λ affects the estimated coefficients.

(b) [3%] Following (a), how does this affect the effective complexity of the resulting function?

(c) [2%] In sum, how is regularization used for reducing overfitting and improving generalization?

3. [4%] In principle component analysis (PCA), the sample vectors are projected to a space of lower dimensionality. Explain the criteria used to determine this projection in PCA, i.e., what is minimized or maximized.

4. [4%] Why are evaluation functions used in game play AIs?

5. [4%] What is the difference between generational and steady-state evolutionary algorithms?

6. [6%] Give example mutation and cross-over operators in genetic algorithms where the chromosomes consist of bit strings.

7. [15%]

(a) [3%] Explain what a multi-armed bandit problem is.

The UCB function is commonly used for move selection in such a problem. It is given by

$$UCB(q) = X(q) + c \sqrt{\frac{\log_{10} n}{n(q)}}$$

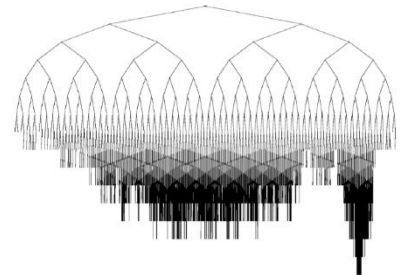
where q is a choice (arm), $X(q)$ is current value (reward ratio) of q , $n(q)$ is the number of times of playing q , n is the number of times of playing all choices, and c is a weighting constant.

(b) [4%] For the two terms in UCB, which one corresponds to exploitation and which to exploration? Briefly explain.

(c) [2%] In the typical 4-stage process of Monte-Carlo Tree Search (MCTS), in which stage is UCB actually used?

(d) [3%] To the right is an example tree formed by MCTS. The tree is imbalanced, unlike the depth-limited tree in standard minimax search. Give a brief explanation.

(e) [3%] Which move (left or right) is to be played from the root state of this MCTS tree? Explain your answer.



8. [6%] Decentralized task allocation is a typical problem for swarm intelligence. A possible form of the probability of individual i to accepting task j is

$$s_j^2 / (s_j^2 + \delta_{ij}^2)$$

where s_j is the stimulus level of task j and δ_{ij} is the response threshold of individual i to task j .

- (a) [3%] Explain, using this expression, how works performed by one individual affect the task choices of the other individuals.
- (b) [3%] Describe a mechanism of introducing "specialization" into task selection.

9. [6%] For ant-colony optimization (ACO),

- (a) [4%] Give reasons of why shorter paths get more pheromone.
- (b) [2%] Give a reason of why it is a good idea to include a decay term (via "pheromone evaporation") of pheromone levels in ACO.

10. [6%] Using these first-order logic sentences:

$$\forall x \text{ King}(x) \wedge \text{Greedy}(x) \Rightarrow \text{Evil}(x)$$

$$\forall y \text{ Greedy}(y)$$

Prove that $\text{Evil}(\text{John})$ from the given fact $\text{King}(\text{John})$ using generalized modus ponens.

11. [4%] The statements "Not everyone likes ice cream" and "There are some who do not like ice cream" are logically equivalent. Express them in first-order logic sentences with appropriate quantifiers.

12. [4%] Explain the difference between model-based and model-free methods in reinforcement learning.

13. [6%] The total reward of a path with state sequence $s_0, s_1, s_2, \dots$ is given by

$$R(s_0) + \gamma R(s_1) + \gamma^2 R(s_2) + \dots$$

Here γ is called the discount factor. Explain its purpose, and how the agent's behavior can differ with larger or smaller value of γ .

14. [7%]

(a) [4%] Explain what a Q function is in reinforcement learning.

(b) [3%] ϵ -greedy is widely used to balance exploitation and exploration in reinforcement learning with Q functions. Explain how it works.

15. [6%] Describe what the term "relational data" means in clustering. Also name/describe two types of clustering methods commonly used for relational data.

16. [8%]

(a) [4%] Sketch an autoencoder.

(b) [2%] Briefly describe the purpose of the "information bottleneck".

(c) [2%] Give a typical loss function for training an autoencoder.